

High-speed based adaptive beamforming handover scheme in LTE-R

ISSN 1751-8628

Received on 19th May 2017

Revised 29th January 2018

Accepted on 22nd February 2018

E-First on 21st May 2018

doi: 10.1049/iet-com.2017.0583

www.ietdl.org

Junhui Zhao^{1,2,3} ✉, Yunyi Liu³, Chuanyun Wang¹, Lei Xiong², Lisheng Fan⁴¹School of Information Engineering, East China Jiaotong University, Nanchang 330013, People's Republic of China²State Key Laboratory of Rail Traffic Control and Safety, Beijing Jiaotong University, Beijing 100044, People's Republic of China³School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing 100044, People's Republic of China⁴School of Computer Science and Educational Software, Guangzhou University, Guangzhou 510006, People's Republic of China

✉ E-mail: junhuizhao@hotmail.com

Abstract: In recent years, high-speed railways (HSRs) are being developed rapidly all over the world because of their convenience, safety, comfort, and other advantages. However, the reliability and security of HSR wireless communication systems have faced severe challenges, such as the Doppler effect, complicated wireless channel model, frequent handovers and so on. The International Union of Railways is pushing the evolution of the global system for mobile communications for railway (GSM-R) internationally. In this study, a high-speed based adaptive beamforming handover scheme is proposed to improve the handover performance for HSR wireless communication systems. When the high-speed train enters the overlapping region, the serving evolved NodeB (eNodeB) and target eNodeB are using beamforming with different gain factors to improve the received signal quality. In addition, this scheme can dynamically adjust the handover hysteresis margins of the reference signal receiving power (RSRP) and the reference signal receiving quality (RSRQ) based on the speed and position. Simulation results have demonstrated that the proposed handover scheme can improve the handover performance by increasing the handover trigger probability and success probability effectively.

1 Introduction

Currently, high-speed railways (HSRs) are gaining increasing popularity for inter-city transportation because of their advantages that include low energy consumption, time savings, high punctuality rate, transport capacity, added comfort, security and so on [1]. HSRs improve the services quality and customer satisfaction on the trip, create enormous social and economic benefits [2]. HSRs have been or will be deployed in an increasing number of countries and regions, such as Japan's Shinkansen, France's Thalys TGV system, Germany's ICE, and China's HSR network [3]. By the end of 2017, China had operated more than 25,000 km of HSRs, accounting for more than 60% of the total HSRs operational mileage in the world. By 2020, China will operate more than 30,000 km of HSRs [4].

However, the existing global system for mobile communications for railway (GSM-R) cannot support the requirements under high mobility scenarios, such as significantly huge data traffic, low end-to-end delay in milliseconds, high reliability and security [5]. Thus, the International Union of Railways (Union Internationale des Chemins de Fer, UIC) has approved long-term evolution for railway (LTE-R) to replace GSM-R as the next-generation railway communication system, which can support the communication requirements in high mobility scenarios [6].

Despite the advantages of the LTE-R, providing a good quality of service (QoS) to almost 1200 passengers in high-speed train at a speed up to 500 km/h, there still have a series of problems to be solved. For example, accurate channel estimation in HSRs environments to guarantee the reliability of communication, advanced signal processing technology to enhance the system spectrum efficiency and energy efficiency, effective handover schemes to reduce latency and outages and avoid communication interruptions and so on [3].

Handover is a key technology which plays an important role in HSR communication systems [7]. It can ensure the continuous connexion of the communication when a train passes through a cell and enhance the validity and reliability of the system. Compared to

the handover schemes in public cellular networks, the special operating environments of HSRs and the high mobility of the trains result in higher requirements for the handover. Many problems remain unsolved, such as a high probability of handover failure, frequent handovers, a large system overhead caused by group handovers and so on [8, 9]. Therefore, it is of vital significance to improve the handover performance and guarantee QoS [10].

In general, the handover trigger scheme is primarily based on the reference signal receiving power (RSRP). When the RSRP of the target evolved NodeB (eNodeB) is higher than that of the source eNodeB for a certain threshold, the handover will be triggered. This scheme is simple and fast to measure and process but it may increase the number of unnecessary handovers [11]. Due to the Doppler effect, shadow fading, multi-path effect, high penetration loss caused by the well-shielded carriages, and other related issues, the RSRP may be very low, which will result in a high handover outage probability [12–14]. Therefore, the speed, distance, bit error rate (BER) and other factors are taken into account in the HSRs scenarios. The joint handover trigger scheme can achieve better system performance and compensate for the defects of a single handover trigger scheme. In [15], the authors proposed a joint handover trigger scheme based on distance and RSRP; the handover was triggered directly when the mobile terminal passed through the midpoint of the overlapping region, and the handover success probability was improved. On this basis, the authors proposed a handover trigger scheme where the handover position was determined by statistical data in [16]. In [17], the authors proposed a joint handover trigger scheme based on RSRP and speed, this scheme used a fast handover scheme to improve the handover success probability when the speed exceeded a threshold. In [18], the handover trigger scheme based on probability of satisfied communication (PoSC) model was proposed, this scheme can reduce the handover numbers and maintain high QoS by considering the effect of received signal strength, co-channel interference and speed on BER. However, this scheme has higher complexity. A joint handover trigger scheme based on the RSRP and the reference signal receiving quality (RSRQ) was proposed in [19]. RSRQ is able to intuitively reflect

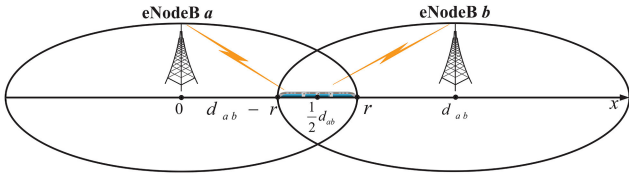


Fig. 1 HSR communication network topology diagram

the link quality of the channel. This scheme thus can select the eNodeB with the best signal quality to perform the handover if the RSRP is not sufficiently high.

The eNodeBs are generally distributed along the rail and the high-speed trains travel along the rail fixedly, these characteristics can facilitate the handover optimisation. Some scholars consider optimising the handover scheme based on the geographical location information. In [20], when the train entered the coverage area of the target eNodeB, the handover trigger condition was adaptively determined according to the speed of the train. The target eNodeB could trigger the handover in advance if the train moved at a high speed. In [21], the direction and speed information provided by global positioning system (GPS) equipment was used to speed up the selection of the target eNodeB and the handover trigger delay was adaptively shortened. A fast handover scheme based on LTE-A network was proposed in [22]. When the train approached a predefined handover position, the network could directly trigger the handover. In [23], the coverage area of the eNodeB was divided into several areas. When the train passed quickly through an area, the network could start the handover earlier. These schemes required network operators to set handover locations in advance, which undoubtedly increased the overhead and cost of the network. And how to inform the handover location to the train was also a problem to be solved, there were two general solutions. One was the network sent the handover location and other information to the train before the train left the station. The other was the eNodeBs sent the handover location to the train by measure control information during the train running. Both solutions will result in additional signalling overhead.

In this paper, we proposed a high-speed based adaptive beamforming handover scheme. When the high-speed train entered the overlapping region, the antennas of the eNodeBs changed its working mode to beamforming mode, and received signal strength of the train could be improved by adjusting the beamforming gain in different areas of the overlapping region. In addition, this scheme could dynamically adjust the handover hysteresis margins of the RSRP and the RSRQ based on the speed and position. The speed and position information is easy to be obtained by devices already implemented on the train, such as GPS, tachometers, accelerators, gyroscopes and so on [24, 25].

The remaining of this paper is organised as follows: The system model and technical details are presented in Section 2. In Section 3, we describe the high-speed based adaptive beamforming handover scheme. Numerical simulation results and analysis are given in Section 4. Finally, Section 5 concludes the entire work.

2 System model, technical details, and design

2.1 System model

The linear coverage topology of the HSR communication network is shown in Fig. 1. The antennas of the eNodeBs can operate in beamforming mode or omnidirectional mode and the beamforming mode is only operated in the overlapping region. We assume that the vertical point of the eNodeB *a* to the rails is the coordinate origin, the train travels from cell *a* to cell *b* along the rails, and the position of the train is *x*. When the antennas of the eNodeBs operate in the omnidirectional mode, the coverage radius is *r*. The distance between the two neighbouring eNodeBs along the rails is *d_{ab}*. Therefore, the length of the overlapping region along the rails is $2r - d_{ab}$.

A Doppler shift in the overlapping region of two neighbouring eNodeBs goes near to steadiness, and can be corrected. In the case of the proposed handover scheme, it is assumed that the Doppler

effect caused by the speed will be removed from the frequency offset compensation technique under the condition of valid channel estimation as well as speed and position information. In addition, the two adjacent eNodeBs are generally not far away and HSRs are mostly developed in an open rural environment where the multi-path effect is very low. Therefore, we do not consider the Doppler effect and multi-path effect in this paper. The train with well shield carriages can cause higher penetration loss of the radio signals (20–35 dB) to/from user equipments (UEs) inside the train. We assume that all UEs in the train access the HSR communication system via the train relay station (TRS) on the top of the train, which provide forwarding information of the signal. The penetration loss can be reduced in this way [26].

In HSR scenarios, a communication interruption is caused if the handover is not triggered at the appropriate time and position. However, if we blindly reduce the handover hysteresis margin, it will result in Ping-Pong effect. Therefore, the handover does not only have to be triggered in time but we also cannot simply reduce the handover hysteresis margin. In order to solve those problems, we proposed a high-speed based adaptive beamforming handover scheme in this paper.

2.2 Beamforming handover scheme

Beamforming is a signal processing technique used in sensor arrays for directional signal transmission or reception that can be understood as spatial filtering [27, 28]. This is achieved by adaptively adjusting the element weights of the antenna arrays so that the signal quality can be optimal under a particular criterion. In HSR scenarios, there is typically only one high-speed train running in a cell, and HSRs are mostly developed in an open rural environment with a strong line of sight. In addition, the eNodeBs and trains can easily calculate direction of arrival (DOA) due to the trains run along the rail fixedly and the speed and position can be predicted. Therefore, DOA beamforming can make good use of these characteristics to realise beamforming. By using beamforming technology, the signals will be transmitted to the running direction of the train, which is beneficial to increase system capacity and reduce inter-cell interference. A geographic location-based beamforming algorithm for HSR was proposed in [29]. The direction of departure (DOD) of beam was calculated according to the geographical location information of eNodeBs and trains.

In HSR scenarios, it is impossible to generate a narrow beam that can accurately track the train because the train's speed is generally about 350 km/h. When the beam generated from the eNodeBs can cover a certain area along the rails, it can serve the train as long as the beamforming gains can be adjusted at the appropriate time and position [30].

When the train enters the overlapping region ($x \in (d_{ab} - r, r)$), the operating mode of the eNodeBs switches to the beamforming model from the omnidirectional model. In order to ensure a smooth handover, in this study, we assume that there are two different beamforming gains G_1 and G_2 ($G_1 > G_2$) at the beamforming mode. The source eNodeB *a* is set to G_1 for the gain of beam 1 and the target eNodeB *b* is set to G_2 for the gain of beam 3. When the train arrives at the middle point of the overlapping region ($x = \frac{1}{2}d_{ab}$), the source eNodeB *a* will reduce the beamforming gain from G_1 to G_2 .

Since the beamforming technique can increase the received signal power from the eNodeBs in a specific direction, the transmit power of the eNodeBs can be appropriately reduced to save energy. When the train moves into the overlapping region, the transmit power of the eNodeB *a* is decreased by G_1 and the transmit power of the eNodeB *b* remains unchanged to maintain minimum communication requirements. We describe some technical details of this scheme in Table 1.

2.3 Dynamic handover trigger scheme

RSRP and RSRQ are key measurement parameters in physical layer. RSRP refers to the received signal power from an eNodeB, which can directly reflect the signal strength at a specific location.

Table 1 Details of beamforming handover mechanism

Beam	eNodeB	Transmit power	Gain of beam	Position
1	a	$P_t - G_1$	G_1	$(d_{ab} - r, \frac{1}{2}d_{ab})$
2	a	$P_t - G_1$	G_2	$(\frac{1}{2}d_{ab}, r)$
3	b	P_t	G_2	$(d_{ab} - r, r)$

Therefore, RSRP is usually used as a primary handover trigger criterion, the RSRP-based handover trigger scheme is commonly used. However, RSRP is unable to be measured in some particular environment due to the interference. At the same time, when the RSRPs of an adjacent eNodeB and the source eNodeB are relatively similar in size, even if the adjacent eNodeB has less interference signals and higher signal-to-interference ratio (SIR), the handover cannot be triggered, which will lead to unreasonable utilisation of channel resources.

RSRQ is similar to the SIR of the link, which is affected by the RSRP and interference signal power. Therefore, RSRQ can objectively reflect the channel load and channel quality. However, RSRQ cannot provide robustness detection when the QoS is degraded. In addition, the system load and measurement strategy have a serious impact on the RSRQ. Therefore, the RSRQ is usually used as an adjunct handover trigger criterion.

Therefore, we use a joint handover trigger scheme based on RSRP and RSRQ to decide whether to trigger the handover, which can guarantee good QoS and improve the handover trigger probability. We define the power of received signal Pr and SIR from the measurement report as the RSRP and RSRQ, respectively. Let Γ_p and Γ_q denote the RSRP and RSRQ hysteresis margins respectively.

The handover trigger scheme procedure is described in Fig. 2. There are two situations that the handover can be triggered. One is that if the Pr received from the target eNodeB is Γ_p dB higher than that from the source eNodeB, the handover is triggered. The other is that if the Pr does not satisfy the trigger condition but the SIR received from the target eNodeB is Γ_q dB higher than that from the source eNodeB, the handover can also be triggered [31].

In the HSR scenarios, a high speed will lead to a rapid change in the path loss. It is unreasonable to adopt the same handover hysteresis margin for different speeds. When the train's speed is high, the time spent in the overlapping region is short and the handover hysteresis margin needs to be lowered to trigger handover in time. In contrast, when the speed is low, the train takes longer to travel in the overlapping region and the handover hysteresis margin has to be improved to avoid the Ping-Pong effect. Therefore, in this study, we dynamically adjust the handover hysteresis margins of the RSRP and RSRQ according to the speed and position of the train.

According to the model proposed in [32], the RSRP hysteresis margin Γ_p is adjusted as follows:

$$\Gamma_p = 7 - \frac{\alpha v \tau}{900}, \quad (1)$$

where v is the train's speed, τ is the time required to complete the handover, and α (dB/km) is the power loss rate of the eNodeB. As the speed increases, Γ_p will be reduced accordingly. When the speed is low, Γ_p will be set a large value to prevent unnecessary handovers. When the speed is high, Γ_p is reduced to increase the handover trigger probability.

When the RSRQ of the source eNodeB is high, it indicates that the QoS of the source eNodeB is good and the RSRQ handover hysteresis margin should be increased to limit the handover. Conversely, if RSRQ of the source eNodeB is low, the RSRQ hysteresis margin should be reduced to facilitate the handover. The RSRQ hysteresis margin Γ_q is adjusted as follows:

$$\Gamma_q = \max \left\{ \Gamma_h \left[\min \left(\frac{\max(\text{SIR} - \Delta_1, 0)}{\Delta_h - \Delta_1}, 1 \right) \right]^k, \Gamma_l \right\}. \quad (2)$$

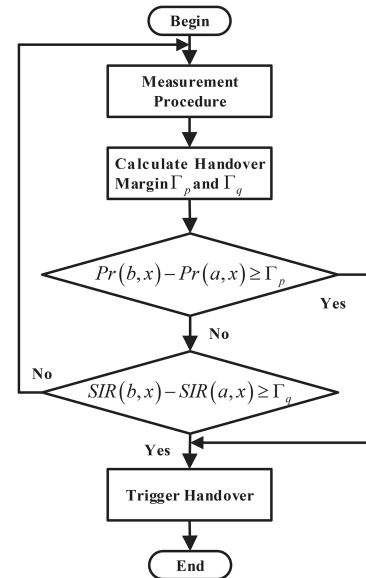


Fig. 2 Handover trigger scheme procedure

where Δ_l and Δ_h denote the lowest and highest levels of link quality, respectively. Γ_l and Γ_h denote the lowest and highest RSRQ hysteresis margins, respectively. k is the velocity factor determining the decrease rate of the RSRQ hysteresis margin, the relationship between k and v is

$$k = \begin{cases} 0.3, & 200 \leq v \leq 300 \\ 0.5, & 300 \leq v \leq 400 \\ 0.7, & 400 \leq v \leq 500 \\ 0.9, & 500 \leq v \end{cases} \quad (3)$$

Therefore, the RSRQ hysteresis margin varies as the location of the train changes. If the SIR is larger than Δ_h , Γ_q is equal to Γ_h , which is used to avoid the Ping-Pong effect. Conversely, if the SIR is lower than Δ_l , Γ_q will be equal to the lowest hysteresis margin Γ_l in order to facilitate the handover.

It is quite obvious that the k denotes descent rate of Γ_q that varies from Γ_h to Γ_l while the RSRQ varies from Δ_h to Δ_l . In addition, because the time it takes for the train to pass through the overlapping region is too short, the higher speed of the train results in a lower Γ_q so that the handover can be triggered in time to avoid communication interruptions.

3 Performance analysis

3.1 Distance calculation

We define that P_t is the transmit power of eNodeBs and d_v is the vertical distance between the eNodeBs and the rails. d_i is the distance between the adjacent co-frequency eNodeBs. In this paper, each eNodeB only considers two adjacent co-frequency eNodeBs. Let eNodeB a_1 and eNodeB a_2 denote the two co-frequency eNodeBs of eNodeB a . Similarly, let eNodeB b_1 and eNodeB b_2 denote the two co-frequency eNodeBs of eNodeB b . The antenna heights of the mobile station and the eNodeB are h_m and h_b , respectively. Therefore, at the position of x , the distance between the train and different eNodeBs can be expressed as

$$\begin{aligned} d_a &= \sqrt{x^2 + d_v^2 + (h_b - h_m)^2} \\ d_b &= \sqrt{(d_{ab} - x)^2 + d_v^2 + (h_b - h_m)^2} \\ d_{a1} &= \sqrt{(d_i + x)^2 + d_v^2 + (h_b - h_m)^2} \\ d_{a2} &= \sqrt{(d_i - x)^2 + d_v^2 + (h_b - h_m)^2} \\ d_{b1} &= \sqrt{(d_i - d_{ab} + x)^2 + d_v^2 + (h_b - h_m)^2} \\ d_{b2} &= \sqrt{(d_i + d_{ab} - x)^2 + d_v^2 + (h_b - h_m)^2}, \end{aligned} \quad (4)$$

where d_a , d_{a_1} and d_{a_2} denote the distances from the train at the position of x to eNodeB a and the two co-frequency eNodeBs. d_b , d_{b_1} and d_{b_2} denote the distances from the train at the position of x to eNodeB b and the two co-frequency eNodeBs.

3.2 Propagation loss model

In most papers, the Okumura–Hata model is typically used to calculate the path loss in HSR scenarios. But the parameter range in Okumura–Hata model does not encompass the 1800 MHz frequency range which is not suitable for HSR. This problem was solved by the COST 231–Hata model, which extends the validity frequency region to the 1500–2000 MHz range. As the HSR is generally built in rural areas, we use the open rural formula to calculate the path loss. Thus the propagation loss is written as [33]:

$$\begin{aligned} PL &= A + B \log(d) + C, \\ A &= 46.3 + 33.9 \log(f_c) - 13.82 \log(h_b) - a(h_m), \\ B &= 44.9 - 6.55 \log(h_b), \\ C &= 0, \\ a(h_m) &= [1.1 \log(f_c) - 0.7] h_m - [1.56 \log(f_c) - 0.8], \end{aligned} \quad (5)$$

where A , B and C are factors that decided by frequency and antenna height, and f_c is given in MHz, d is given in kilometre, h_b and h_m are given in metre.

3.3 Traditional handover scheme

Let $eNBC = [a, b, a_1, a_2, b_1, b_2]$ be the basis set of eNodeBs. Therefore, the strength of the received signal $Pr(m, x)$ from eNodeB m at the position of x is

$$\begin{aligned} Pr(m, x)[\text{dBm}] \\ &= Pr[\text{dBm}] - PL(m, d_m)[\text{dB}] - \varepsilon(0, \sigma_m)[\text{dB}], \quad m \in eNBC, \end{aligned} \quad (6)$$

where $PL(m, d_m)$ is the propagation loss from eNodeB m at the position of x , and $\varepsilon(0, \sigma_m)$ denotes the shadow fading that obeys a Gaussian distribution with a mean of 0 and a standard deviation of σ_m .

The strength of the interference signals that received from the co-frequency eNodeBs are

$$I(a, x)[\text{dBm}] = 10 \log_{10} \left(10^{(Pr(a_1, x))/10} + 10^{(Pr(a_2, x))/10} \right), \quad (7)$$

$$I(b, x)[\text{dBm}] = 10 \log_{10} \left(10^{(Pr(b_1, x))/10} + 10^{(Pr(b_2, x))/10} \right). \quad (8)$$

The signal quality received from eNodeB a and eNodeB b at the position of x is

$$\text{SIR}(a, x)[\text{dB}] = Pr(a, x)[\text{dBm}] - I(a, x)[\text{dBm}], \quad (9)$$

$$\text{SIR}(b, x)[\text{dB}] = Pr(b, x)[\text{dBm}] - I(b, x)[\text{dBm}]. \quad (10)$$

The handover trigger scheme depends on the RSRP and RSRQ. If the Pr from the target eNodeB is Γ dB higher than that from the source eNodeB, the handover will be triggered. Whereas if the Pr does not satisfy the handover trigger condition but the SIR from the target eNodeB is Γ dB higher than that from the source eNodeB, then the handover will be executed. Therefore, the handover trigger probability of the traditional handover scheme is

$$\begin{aligned} P(x)_{\text{handover}} &= P[Pr(b, x) - Pr(a, x) \geq \Gamma] \\ &\quad + P[Pr(b, x) - Pr(a, x) < \Gamma] \\ &\quad \times P[\text{SIR}(b, x) - \text{SIR}(a, x) \geq \Gamma], \end{aligned} \quad (11)$$

(see (12)) where Γ is the general value of the handover hysteresis margin. The detailed calculation of (11) is shown in (12) at the bottom of this page. $Q(x)$ is the probability that a normal (Gaussian) random variable will obtain a value larger than x standard deviations above the mean which is defined as

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty \exp\left(-\frac{t^2}{2}\right) dt. \quad (13)$$

Next, another significant index to verify the performance of the handover scheme is the handover outage probability. This represents the probability that the received signal quality is very poor and causes a communication interruption. In this paper, we assume that the outage occurs if the SIR is smaller than Υ dB. Thus, the handover outage probability are denoted as:

$$\begin{aligned} P(a, x)_{\text{out}} &= P[\text{SIR}(a, x) < \Upsilon] \\ &= P[Pr(a, x) - I(a, x) < \Upsilon] \\ &= P[Pt - PL(a, d_a) - \varepsilon(0, \sigma_a) - I(a, x) < \Upsilon] \\ &= P[\varepsilon(0, \sigma_a) > Pt - PL(a, d_a) - I(a, x) - \Upsilon] \\ &= Q\left(\frac{Pt - PL(a, d_a) - I(a, x) - \Upsilon}{\sigma_a}\right), \\ P(b, x)_{\text{out}} &= Q\left(\frac{Pt - PL(b, d_b) - I(b, x) - \Upsilon}{\sigma_b}\right). \end{aligned} \quad (14)$$

In the handover scheme, we define the handover success probability when the following three conditions are satisfied. First, there are no communication interruptions occur prior to triggering

$$\begin{aligned} P[Pr(b, x) - Pr(a, x) \geq \Gamma] &= P[-PL(b, d_b) + PL(a, d_a) - \varepsilon(0, \sigma_b) + \varepsilon(0, \sigma_a) \geq \Gamma] \\ &= P[\varepsilon(0, \sigma_a) \geq \Gamma + PL(b, d_b) - PL(a, d_a) + \varepsilon(0, \sigma_b)] \\ &= P[\varepsilon(0, \sigma_a) \geq \Gamma + PL(b, d_b) - PL(a, d_a) + \varepsilon_0 | \varepsilon(0, \sigma_b) = \varepsilon_0] \\ &\quad \times P[\varepsilon(0, \sigma_b) = \varepsilon_0] \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_b^2}} Q\left(\frac{\Gamma + PL(b, d_b) - PL(a, d_a) + \varepsilon_0}{\sigma_a}\right) \\ &\quad \times \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \\ P[Pr(b, x) - Pr(a, x) < \Gamma] &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_b^2}} \left[1 - Q\left(\frac{\Gamma + PL(b, d_b) - PL(a, d_a) + \varepsilon_0}{\sigma_a}\right)\right] \\ &\quad \times \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \\ P[\text{SIR}(b, x) - \text{SIR}(a, x) \geq \Gamma] &= \int_{-\infty}^{\infty} Q\left(\frac{\Gamma + PL(b, d_b) - PL(a, d_a) + I(b, x) - I(a, x) + \varepsilon_0}{\sigma_a}\right) \\ &\quad \times \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \end{aligned} \quad (12)$$

a handover. Second, the handover is triggered and successfully executed in a suitable position. Third, there are no communication interruptions and the Ping-Pong handover phenomenon occur after the handover is executed. Therefore, the handover success probability is denoted as follows:

$$P(x)_{\text{success}} = \prod_{z=0}^x [1 - P(a, z)_{\text{out}}] \times P(x)_{\text{handover}} \cdot \prod_{z=x}^{d_{ab}} [1 - P(b, z)_{\text{out}}]. \quad (15)$$

3.4 Proposed handover scheme

The proposed scheme in this paper focuses on the handover performance in the overlapping region. According to Table 1 in Section 2, the source and target eNodeBs make different adjustments to the respective beamforming gains when the train crosses the overlapping region; thus, the strength of the received signal from eNodeB a is

$$\begin{aligned} Pr(a, x)_{\text{BF}} &= Pt - G_1 - PL(a, d_a) + G_1 - \varepsilon(0, \sigma_a) \\ &= Pt - PL(a, d_a) - \varepsilon(0, \sigma_a), \\ x &\in \left(d_{ab} - r, \frac{1}{2}d_{ab}\right), \end{aligned} \quad (16)$$

$$\begin{aligned} Pr(a, x)_{\text{BF}} &= Pt - G_1 - PL(b, d_b) + G_2 - \varepsilon(0, \sigma_a) \\ &= Pt - PL(b, d_b) + G_2 - G_1 - \varepsilon(0, \sigma_a), \\ x &\in \left(\frac{1}{2}d_{ab}, r\right). \end{aligned} \quad (17)$$

Similarly, the received signal strength from eNodeB b is

$$\begin{aligned} Pr(b, x)_{\text{BF}} &= Pt - PL(b, d_b) + G_2 - \varepsilon(0, \sigma_b), \\ x &\in (d_{ab} - r, r). \end{aligned} \quad (18)$$

The use of the beamforming technique reduces the transmit power and interference of the source eNodeB. Let β represent the interference elimination factor, which is the degree of interference reduction. Then the received interference signals from the co-frequency eNodeBs can be denoted as

$$I(a, x)_{\text{BF}}[\text{dBm}] = 10\log_{10}\left[\beta\left(10^{\frac{Pr(a_1, x)}{10}} + 10^{\frac{Pr(a_2, x)}{10}}\right)\right], \quad (19)$$

$$I(b, x)_{\text{BF}}[\text{dBm}] = 10\log_{10}\left[\beta\left(10^{\frac{Pr(b_1, x)}{10}} + 10^{\frac{Pr(b_2, x)}{10}}\right)\right]. \quad (20)$$

The quality of received signals from eNodeB a and b is

$$\text{SIR}(a, x)_{\text{BF}}[\text{dB}] = Pr(a, x)_{\text{BF}}[\text{dBm}] - I(a, x)_{\text{BF}}[\text{dBm}], \quad (21)$$

$$\text{SIR}(b, x)_{\text{BF}}[\text{dB}] = Pr(b, x)_{\text{BF}}[\text{dBm}] - I(b, x)_{\text{BF}}[\text{dBm}]. \quad (22)$$

Therefore, according to the dynamic handover trigger scheme, the handover trigger probability is

$$\begin{aligned} P(x)_{\text{BF_handover}} &= P[Pr(b, x)_{\text{BF}} - Pr(a, x)_{\text{BF}} \geq \Gamma_p] \\ &\quad + P[Pr(b, x)_{\text{BF}} - Pr(a, x)_{\text{BF}} < \Gamma_p] \\ &\quad \times P[\text{SIR}(b, x)_{\text{BF}} - \text{SIR}(a, x)_{\text{BF}} \geq \Gamma_q]. \end{aligned} \quad (23)$$

where Γ_p and Γ_q are the dynamic handover hysteresis margins calculated by (1) and (2). The detailed calculations of (23) are shown in (24) and (25).

When $x \in (d_{ab} - r, \frac{1}{2}d_{ab})$, (see (24)). When $x \in (\frac{1}{2}d_{ab}, r)$,

$$\begin{aligned} &P[Pr(b, x)_{\text{BF}} - Pr(a, x)_{\text{BF}} \geq \Gamma_p] \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_b^2}} Q\left(\frac{\Gamma_p + PL(b, d_b) - PL(a, d_a) - G_1 + \varepsilon_0}{\sigma_a}\right) \\ &\quad \times \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \\ &P[Pr(b, x)_{\text{BF}} - Pr(a, x)_{\text{BF}} < \Gamma_p] \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_b^2}} \left[1 - Q\left(\frac{\Gamma_p + PL(b, d_b) - PL(a, d_a) - G_1 + \varepsilon_0}{\sigma_a}\right)\right] \\ &\quad \times \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \\ &P[\text{SIR}(b, x)_{\text{BF}} - \text{SIR}(a, x)_{\text{BF}} \geq \Gamma_q] \\ &= \int_{-\infty}^{\infty} Q \\ &\quad \times \left(\frac{\Gamma_q + PL(b, d_b) - PL(a, d_a) + I(b, x)_{\text{BF}} - I(a, x)_{\text{BF}} - G_1 + \varepsilon_0}{\sigma_a}\right) \\ &\quad \times \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \end{aligned} \quad (25)$$

$$\begin{aligned} &P[Pr(b, x)_{\text{BF}} - Pr(a, x)_{\text{BF}} \geq \Gamma_p] \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_b^2}} Q\left(\frac{\Gamma_p + PL(b, d_b) - PL(a, d_a) - G_2 + \varepsilon_0}{\sigma_a}\right) \\ &\quad \times \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \\ &P[Pr(b, x)_{\text{BF}} - Pr(a, x)_{\text{BF}} < \Gamma_p] \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_b^2}} \left[1 - Q\left(\frac{\Gamma_p + PL(b, d_b) - PL(a, d_a) - G_2 + \varepsilon_0}{\sigma_a}\right)\right] \\ &\quad \times \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \\ &P[\text{SIR}(b, x)_{\text{BF}} - \text{SIR}(a, x)_{\text{BF}} \geq \Gamma_q] \\ &= \int_{-\infty}^{\infty} Q \\ &\quad \times \left(\frac{\Gamma_q + PL(b, d_b) - PL(a, d_a) + I(b, x)_{\text{BF}} - I(a, x)_{\text{BF}} - G_2 + \varepsilon_0}{\sigma_a}\right) \\ &\quad \times \frac{1}{\sqrt{2\pi\sigma_b^2}} \exp\left(-\frac{\varepsilon_0^2}{2\sigma_b^2}\right) d\varepsilon_0 \end{aligned} \quad (24)$$

Table 2 Simulation parameters

Parameter	Value	Parameter	Value
f_c	2 GHz	σ_m	4 dB
P_t	43 dBm	Γ	3 dB
h_b	30 m	Υ	-35 dB
h_m	1.5 m	β	0.01
r	3 km	Γ_h	20 dB
d_{ab}	4.8 km	Γ_l	3 dB
d_i	14.4 km	Δ_h	60 dB
d_v	30 m	Δ_l	15 dB
α	19.8 dB/km	G_1	5 dB
τ	500 ms	G_2	3 dB

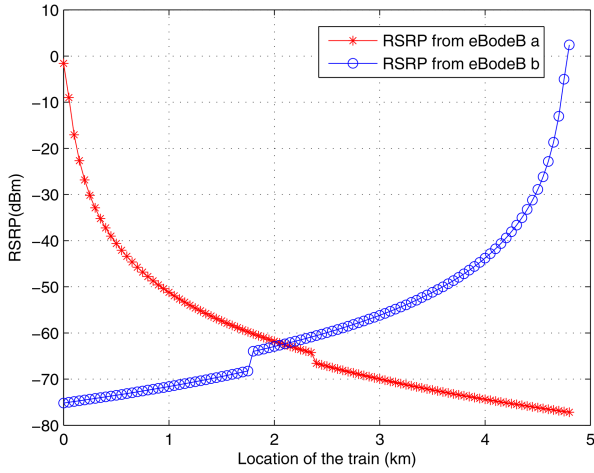


Fig. 3 Relationship between the RSRP and the location

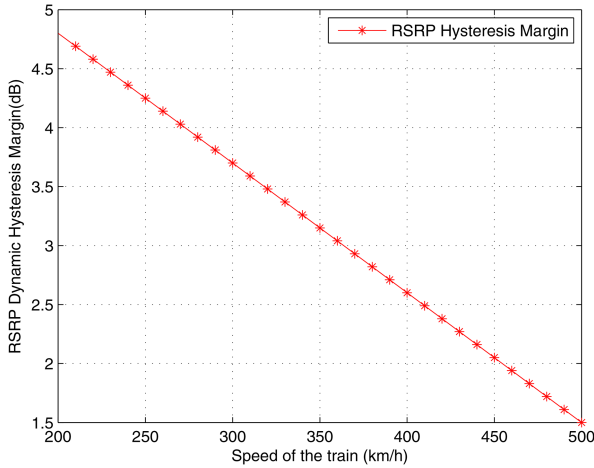


Fig. 4 RSRP dynamic hysteresis margin

Similar to the traditional handover scheme, the handover outage probabilities in our proposed handover scheme are denoted as

$$\begin{aligned}
 P(a, x)_{\text{BF_out}} &= Q\left(\frac{P_t - \text{PL}(a, d_a) - I(a, x) - \Upsilon}{\sigma_a}\right), \\
 x &\in \left(d_{ab} - r, \frac{1}{2}d_{ab}\right). \\
 P(a, x)_{\text{BF_out}} &= Q\left(\frac{P_t - \text{PL}(a, d_a) - I(a, x) - \Upsilon + G_2 - G_1}{\sigma_a}\right), \quad (26) \\
 x &\in \left(\frac{1}{2}d_{ab}, r\right). \\
 P(b, x)_{\text{BF_out}} &= Q\left(\frac{P_t - \text{PL}(b, d_b) - I(b, x) - \Upsilon + G_1}{\sigma_b}\right), \\
 x &\in (d_{ab} - r, r).
 \end{aligned}$$

Therefore, the handover success probability is denoted as

$$\begin{aligned}
 P(x)_{\text{BF_success}} &= \prod_{z=0}^x [1 - P(a, z)_{\text{BF_out}}] \\
 &\times P(x)_{\text{BF_handover}} \cdot \prod_{z=x}^{d_{ab}} [1 - P(b, z)_{\text{BF_out}}]. \quad (27)
 \end{aligned}$$

4 Numerical simulation and analysis

In this paper, we used the Matlab as the simulation tool to verify the performance of our proposed handover scheme. The simulation parameters in this scheme are shown in Table 2.

Fig. 3 shows the relationship between the RSRP and the location of the train where the line with asterisk denotes the RSRP received from the source eNodeB a and the line with circle denotes the RSRP received from the target eNodeB b . Because of the beamforming technique, the eNodeB a and the eNodeB b adjust the different beam gain factors when the train enters the overlapping region. Therefore, the RSRP received from the target eNodeB b is increased by 3 dBm at the position of $x = 1.8$ km and the RSRP received from the source eNodeB a is reduced by 2 dBm at the position of $x = 2.4$ km. In this way, we can improve the signal quality at the cell edge to ensure the smooth progress of the handover and avoid communication interruptions.

Fig. 4 illustrates the relationship between the RSRP dynamic hysteresis margin and the speed of the train. The traditional RSRP-based handover trigger scheme usually takes 3 dB as the fixed hysteresis margin, which is not suitable for HSR scenarios. As shown in Fig. 4, the RSRP hysteresis margin is 4.8 dB when the speed is 200 km/h. As the speed increases, the RSRP hysteresis margin is reduced accordingly to increase the handover trigger probability and improve the handover success probability.

Fig. 5 illustrates the relationship between the RSRQ dynamic hysteresis margin, speed, and the location of the train. It can be seen that the faster the speed, the faster the RSRQ hysteresis margin decreases; thus, the handover can be triggered faster at a higher speed. Meanwhile, the RSRQ hysteresis margin is decreasing as the distance increases but it will remain constant when it is reduced to 3 dB. This is because we set the minimum hysteresis margin in the previously defined scheme and this can effectively avoid the occurrence of the Ping-Pong effect.

Fig. 6 shows the numerical simulation results of the handover trigger probability with different schemes. As the handover hysteresis margins of the other schemes are fixed, it can be concluded that the handover trigger probability is almost unchanged when the speed is different. The dynamic handover trigger scheme proposed in this paper can trigger the handover more quickly according to the decision conditions. At the position of $x = 2.45$ km, the handover trigger probabilities of the traditional scheme and the beamforming and positioning-assisted (BPA) handover scheme in [30] are 51.2 and 72.7%, respectively. At this point, the handover trigger probability of the proposed scheme shows a significant improvement due to the beam gain adjustment and the handover trigger probability is increased to a minimum of 85.7% when the speed is 200 km/h. At the edge of the source cell ($x = 3$ km), the handover trigger probability of the traditional scheme is 98.6%. However, the handover trigger probability can achieve the same value at $x = 2.8$ km in the proposed scheme. Because of the dynamic handover trigger scheme, the proposed scheme achieves a good adaptability in the HSR scenarios.

Fig. 7 shows the numerical simulation results of the handover success probability with different schemes. The handover success probability of the proposed scheme is higher than the traditional handover scheme and BPA handover scheme. At the position of $x = 2.8$ km, the handover success probabilities of the traditional scheme and BPA handover scheme are 90.7% and 96.0% respectively, while the value for the proposed scheme is 97.9%. This is because the handover can be triggered more quickly and accurately due to the dynamic adjustment of the handover hysteresis margins and the beamforming technique. Thus, there is sufficient time to complete the handover.

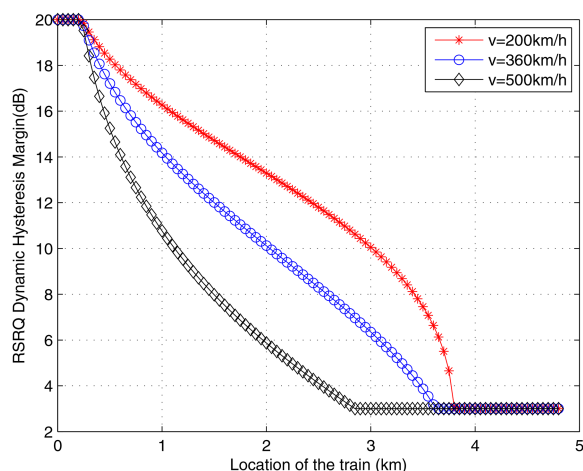


Fig. 5 RSRQ dynamic hysteresis margin

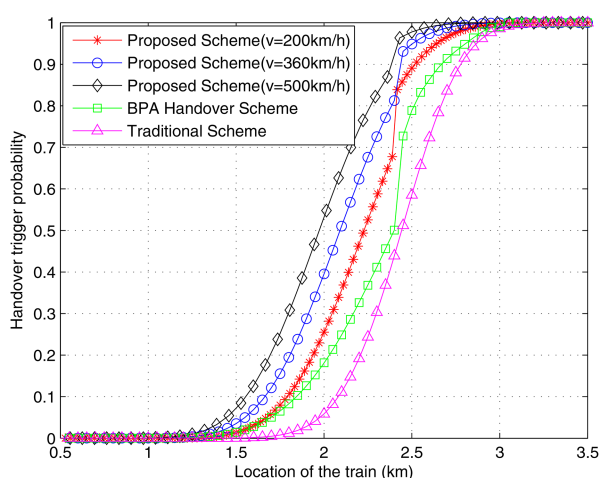


Fig. 6 Handover trigger probability

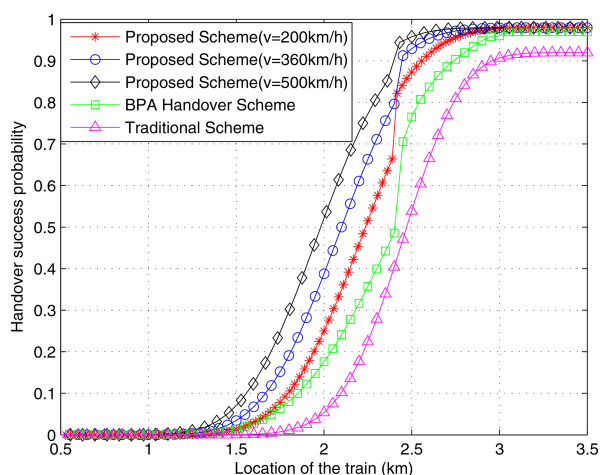


Fig. 7 Handover success probability

5 Conclusions

The HSR communication system is faced with the challenges of reliability and efficiency due to the high mobility. Taking into account the evolution of the GSM-R to the LTE-R, we use the beamforming technique by adjusting the beamforming gain at a right time and place to improve the received signal quality. At the same time, aiming at improving the handover performance, the handover hysteresis margins dynamic adjustment scheme of the RSRP and RSRQ is given by analysing the influence of high speed on the handover trigger decision. On this basis, a high-speed based adaptive beamforming handover scheme is proposed in this paper. We have compared and analysed the performances of the other

handover schemes and the proposed scheme. By applying the beamforming technique and the dynamic handover trigger scheme, simulation results show that the handover trigger probability and handover success probability is significantly improved in HSR scenarios.

6 Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (grant nos. 61661021, 61471031), in part by the Open Research Fund of National Mobile Communications Research Laboratory, Southeast University (grant no. 2017D14), in part by the State Key Laboratory of Rail Traffic Control and Safety (contract no. RCS2017K009), in part by Science and Technology Project of Jiangxi Provincial Transport Bureau (grant no. 2016D0037), in part by Science and Technology Program of Jiangxi Province (grant nos. 20172BCB22016, 20171BBE50057).

7 References

- [1] Song, H., Fang, X., Yan, L.: 'Handover scheme for 5G C/U plane split heterogeneous network in high-speed railway', *IEEE Trans. Veh. Technol.*, 2014, **63**, (9), pp. 4633–4646
- [2] He, R., Ai, B., Wang, G., *et al.*: 'High-speed railway communications: from GSM-R to LTE-R', *IEEE Veh. Technol. Mag.*, 2016, **11**, (3), pp. 49–58
- [3] Fan, P., Zhao, J., Chih Lin, I.: '5G high mobility wireless communications: challenges and solutions', *China Commun.*, 2016, **13**, (Supplement 2), pp. 1–13
- [4] Wang, J., Zhu, H., Gomes, N.J.: 'Distributed antenna systems for mobile communications in high speed trains', *IEEE J. Sel. Areas Commun.*, 2012, **30**, (4), pp. 675–683
- [5] Ai, B., Cheng, X., Kürner, T., *et al.*: 'Challenges toward wireless communications for high-speed railway', *IEEE Trans. Intell. Transp. Syst.*, 2014, **15**, (5), pp. 2143–2158
- [6] Karimi, O.B., Liu, J., Wang, C.: 'Seamless wireless connectivity for multimedia services in high speed trains', *IEEE J. Sel. Areas Commun.*, 2012, **30**, (4), pp. 729–739
- [7] Li, Y., Cao, B., Wang, C.: 'Handover schemes in heterogeneous LTE networks: challenges and opportunities', *IEEE Wirel. Commun.*, 2016, **23**, (2), pp. 112–117
- [8] Tian, L., Li, J., Huang, Y., *et al.*: 'Seamless dual-link handover scheme in broadband wireless communication systems for high-speed rail', *IEEE J. Sel. Areas Commun.*, 2012, **30**, (4), pp. 708–718
- [9] Liu, Z., Fan, P.: 'An effective handover scheme based on antenna selection in ground-train distributed antenna systems', *IEEE Trans. Veh. Technol.*, 2014, **63**, (7), pp. 3342–3350
- [10] Jansen, T., Balan, I., Stefanski, S., *et al.*: 'Weighted performance based handover parameter optimization in LTE'. Proc. IEEE 73rd Vehicular Technology Conf. (VTC), Yokohama, Japan, May 2011, pp. 1–5
- [11] Zonoozi, M.M., Dassanayake, P.: 'User mobility modeling and characterization of mobility patterns', *IEEE J. Sel. Areas Commun.*, 1997, **15**, (7), pp. 1239–1252
- [12] Li, W., Zhang, C., Duan, X., *et al.*: 'Performance evaluation and analysis on group mobility of mobile relay for LTE advanced system'. Proc. IEEE 76th Vehicular Technology Conf. (VTC) Fall, Québec City, Canada, September 2012, pp. 1–5
- [13] Zhao, J., Wang, S., Liu, X., *et al.*: 'Geometry-based stochastic modeling for non-stationary high-speed train MIMO channels'. Proc. IEEE 84th Vehicular Technology Conf. (VTC) Fall, Montréal, Canada, September 2016, pp. 1–5
- [14] Pan, M.S., Lin, T.M., Chen, W.T.: 'An enhanced handover scheme for mobile relays in LTE-A high-speed rail networks', *IEEE Trans. Veh. Technol.*, 2015, **64**, (2), pp. 743–756
- [15] Luo, W., Fang, X., Cheng, M., *et al.*: 'An optimized handover trigger scheme in LTE systems for high-speed railway'. Proc. 2011 5th Int. Workshop on Signal Design and its Applications in Communications (IWSDA), Guilin, China, October 2011, 193–196
- [16] Yang, Y., Yu, P., Li, W.: 'Handover self-optimization mechanism based on velocity for cellular networks'. Proc. IEEE 7th Chinacom, Kun Ming, China, August 2012, pp. 606–610
- [17] Bhattacharya, S., De, D.: 'An enhanced handover strategy for GSM-R technology'. Proc. IEEE 3rd Int. Conf. on Computer and Communication Technology (ICCCCT), Allahabad, India, November 2012, pp. 227–232
- [18] Sun, B., Feng, Q.K., Jin, X.Y., *et al.*: 'Handoff algorithm and PoSC model under high-speed railway environment', *J. Zhejiang Univ. Eng. Sci.*, 2012, **46**, (9), pp. 1553–1559
- [19] Yuhan, C.: 'Simulation research on handover algorithm for TD-LTE in high speed rail based on NS2' (Beijing University of Posts and Telecommunications, Beijing, China, 2012)
- [20] Aguado, M., Jacob, E., Astorga, J., *et al.*: 'The cross layer RMPA handover: a reliable mobility pattern aware handover strategy for broadband wireless communication in a high-speed railway domain', *EURASIP J. Wirel. Commun. Netw.*, 2012, **2012**, (1), p. 298
- [21] Chang, F.M., Wang, H.L., Hu, S.Y., *et al.*: 'An efficient handover mechanism by adopting direction prediction and adaptive time-to-trigger in LTE networks'. Proc. Int. Conf. on Computational Science and its Applications (ICCSA), Ho Chi Minh City, Vietnam, June 2013, pp. 270–280

- [22] Huang, Q., Zhou, J., Tao, C., *et al.*: 'Mobile relay based fast handover scheme in high-speed mobile environment'. Proc. IEEE 76th Vehicular Technology Conf. (VTC) Fall, Québec City, Canada, September 2012, pp. 1–6
- [23] Huang, Y.S., Leu, F.Y., Liu, J.C., *et al.*: 'A handover scheme for LTE wireless networks under the assistance of GPS'. Proc. IEEE 8th Broadband, Wireless Computing, Communication and Applications (BWCCA), Compiègne, France, October 2013, pp. 399–403
- [24] Ai, B., Guan, K., Rupp, M., *et al.*: 'Future railway services-oriented mobile communications network', *IEEE Commun. Mag.*, 2015, **53**, (10), pp. 78–85
- [25] Pascoe, R., Eichorn, T.: 'What is communication-based train control?', *IEEE Veh. Technol. Mag.*, 2009, **4**, (4), pp. 16–21
- [26] Junhui, Z., Yunyi, L., Yi, G., *et al.*: 'A dual-link soft handover scheme for C/U plane split network in high-speed railway', *IEEE Access*, 2018, **6**, pp. 12473–12482
- [27] Roh, W., Seol, J.Y., Park, J., *et al.*: 'Millimeter-wave beamforming as an enabling technology for 5G cellular communications: theoretical feasibility and prototype results', *IEEE Commun. Mag.*, 2014, **52**, (2), pp. 106–113
- [28] Cui, Y., Fang, X., Yan, L.: 'Hybrid spatial modulation beamforming for mmWave railway communication systems', *IEEE Trans. Veh. Technol.*, 2016, **65**, (12), pp. 9597–9606
- [29] Wang, C., Xu, R., You, J., *et al.*: 'Location assistant beamforming for high speed railway'. Proc. IEEE 8th Int. Wireless Communications and Mobile Computing Conf. (IWCMC), Limassol, Cyprus, August 2012, pp. 1108–1112
- [30] Cheng, M., Fang, X., Luo, W.: 'Beamforming and positioning-assisted handover scheme for long-term evolution system in high-speed railway', *IET Commun.*, 2012, **6**, (15), pp. 2335–2340
- [31] Xia, Y., Fang, X., Luo, W., *et al.*: 'Coordinated of multi-point and bi-casting joint soft handover scheme for high-speed rail', *IET Commun.*, 2014, **8**, (14), pp. 2509–2515
- [32] Zhenzhen, H.: 'The research on handover technology under high-speed mobile communication system based on LTE-R' (Central South University, Changsha, China, 2013)
- [33] Molisch, A.F.: 'Wireless communications' (John Wiley & Sons, Hoboken, USA, 2012)